INCH-POUND

MIL-DTL-24643/63
1 March 2007

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, TYPE LSYSGU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable type is specifically designed for single circuit (see Figure 1), three-phase, wye connected (with neutral wire), 60 or 400 Hertz, electrical power distribution. The designation Y represents the wye connection. Using two conductors per phase reduces the series inductance to 50 percent of the single conductor per phase value. In addition, the symmetrical spacing of the phase conductors from the neutral wire provides for balanced voltage drop when three-phase loads are applied.

REQUIREMENTS:

Qualification Required.

Construction (Watertight with Circuit Integrity)

First	-	Copper conductor, uncoated (see Table I for size).
Second	-	Separator may be used at the manufacturer's option to ensure that the insulation is easily stripped off of the conductor.
Third	-	Extruded silicone rubber, or silicone rubber-glass tape insulation at the manufacturer's option (see Table I for diameter). Refer to Table II for identification code.
Fourth	-	Seven conductors cabled together with a left-hand lay of six conductors around the center conductor (see Figure 2 for cross-sectional view). The length of lay shall not exceed 24 times the pitch diameter. Fillers shall be employed to obtain a well-rounded assembly.
Fifth	-	Optional binder tape applied helically with overlap.
Sixth	-	Cross-linked polyolefin jacket (see Table I for thickness).

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TABLE I. LSYSGU sizes and characteristics.

Military Part No. M24643/63	Conductor							Cable		
	AWG	Circular MILS	Stranding	Diameter Over Insulation (min.)	Amps per Phase (2 conductors)		Ω/1000 ft @ 25 °C (max.)	Jacket Thickness (min.)	Overall Diameter (max.)	Weight (lbs/ft)
					40 °C ambient	50 °C ambient				
-01UN	16	2580	7 X 0.0192	0.125	18	16	4.300	0.040	0.535	0.188
-02UN	14	4110	7 X 0.0242	0.140	29	27	2.680	0.040	0.580	0.255
-03UN	12	6530	7 X 0.0305	0.160	39	36	1.900	0.040	0.640	0.314
-04UN	10	10380	7 X 0.0385	0.187	62	58	1.125	0.040	0.730	0.420
-05UN	8	16510	19 X 0.0295	0.275	84	77	0.790	0.050	1.000	0.725
-06UN	4	41740	19 X 0.0469	0.325	145	133	0.287	0.050	1.170	1.400
-07UN	2	66360	19 X 0.0591	0.365	193	178	0.203	0.060	1.310	2.000
-08UN	1	83690	37 X 0.0476	0.407	237	218	0.145	0.060	1.450	2.400
-09UN	0	105600	61 X 0.0416	0.453	272	254	0.120	0.060	1.600	3.000
-10UN	00	133100	61 X 0.0467	0.510	320	294	0.0888	0.065	1.780	3.750
-11UN	000	167800	61 X 0.0524	0.560	376	346	0.0674	0.065	1.880	4.500
Note: All dimensions are in inches unless otherwise specified.										

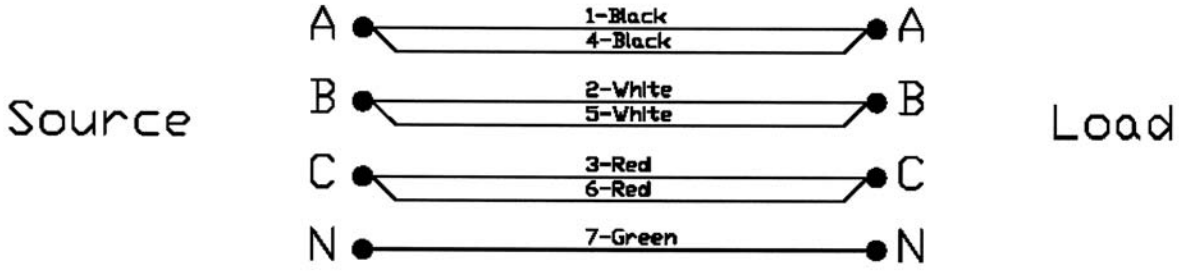


FIGURE 1. LSYSGU single circuit user diagram.

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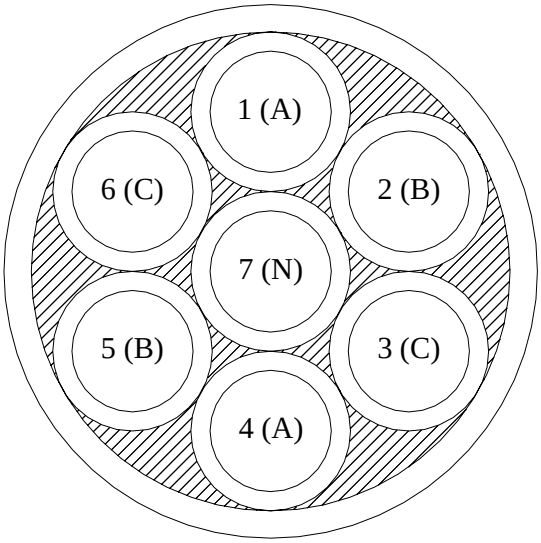


FIGURE 2. LSYSGU cross-section design.

TABLE II. LSYSGU identification code and color.

Conductor #	Identification Code	Color
1	A	Black
2	B	White
3	C	Red
4	A	Black
5	B	White
6	C	Red
7	N	Green

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EXAMINATION AND TESTS:	
Basic Electricals:	
Conductor resistance (ohms/1000 feet at 25 °C, max.)	see Table II
Voltage withstand (root mean square, min.)	
Sizes 16 through 10	3000 VAC
Sizes 8 and larger	5000 VAC
Insulation resistance (megohms/1000 feet, min.)	100
Conductor continuity	No Failure
Group A:	
Visual and dimensional	No Failure
Watertightness (see MIL-DTL-24643 for limits of water leakage)	No Failure
Group B:	
Cross-linked proof test (percent, max.)	
Insulation	50
Jacket (when tested at 200±5 °C)	50
Tear (pounds per inch thickness, min.) (ASTM D470)	35
Gas flame (1 hour)	No Failure
Drip (95±1 °C)	Zero
Physicals (unaged)	
Insulation	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	250
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Cold bending cable (mandrel is 12 times the outer diameter)	No Damage
Group C:	
Physicals (aged)	
Jacket (cable)	
Air oven	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60

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Hot oil immersion	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Shrinkage	No Failure
Heat distortion (percent of unaged, max.)	30
Permanence of printing (jacket) (cycles, min.)	125
Group D:	
Flame propagation (cable)	No Failure
Qualification Inspection:	
Qualification inspection shall include basic electricals, groups A, B, C, and D, plus the following:	
Accelerated service (1.8 times the amps per phase @50 °C ambient)	No Failure
Aging and compatibility (cable) (125±5 °C)	No Failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Gas flame (3 hours)	No Failure
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index, max.	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, max.	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – Type A (hardness, min.)	80

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Weathering (jacket)	No Failure

UNIT ORDERING LENGTHS: 500 feet min.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2006-067)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – IS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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